1 Bases

We have seen that if $\{v_1,\dots,v_m\}$ is a spanning set for V, then any $v \in V$ can be written as

$$v = \alpha_1 v_1 + \cdots + \alpha_m v_m,$$

for some $\alpha_1,\dots,\alpha_m \in F$. We have also seen that there is a unique choice of these scalars if and only if $\{v_1,\dots,v_m\}$ is linearly independent. Thus, in the case where $\{v_1,\dots,v_m\}$ is both linearly independent and spans V, then every vector $v \in V$ can be represented uniquely as a linear combination of $\{v_1,\dots,v_m\}$. In this case, we say that $\{v_1,\dots,v_m\}$ is a *basis* of V.

Definition 1.1 (Basis)**.** A *basis* of V is a subset of V which is linearly independent and spans V.

Example 1.2. We have seen that $\{(1,0,\dots,0),(0,1,0,\dots,0),\dots,(0,\dots,0,1)\}$ is linearly independent and spans F^n, and is therefore a basis of F^n. It is called the *standard basis* of F^n.

Example 1.3. We have seen that $\{1,t,t^2,\dots,t^m\}$ is linearly independent and spans $P_m(F)$, the vector space of polynomial of degree at most m with coefficients in F. This is called the *standard basis* of $P_m(F)$.

Exercise 1.4. Prove that $\{(1,2),(3,5)\}$ is a basis of F^2.

Remark 1.5. We saw in Example 1.2 that $\{(1,0),(0,1)\}$ is also a basis of F^2, so the previous problem shows, in particular, that bases are not unique.

Exercise 1.6. Let U be the subspace of F^5 defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in F^5 : x_1 = 3x_2 \text{ and } x_3 = 7x_4\}.$$

Find a basis of U.

Exercise 1.7. Let U be the subspace of F^5 defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in F^5 : 6x_1 = x_2 \text{ and } x_3 + 2x_4 + 3x_5 = 0\}.$$

Find a basis of U.

Exercise 1.8. Prove or give a counterexample: If $\{p_0, p_1, p_2, p_3\} \subset P_3(F)$ and none of the polynomials p_0, p_1, p_2, p_3 has degree 2, then $\{p_0, p_1, p_2, p_3\}$ is not a basis of $P_3(F)$.

Exercise 1.9. Suppose $\{v_1, v_2, v_3, v_4\}$ is a basis of V. Prove that $\{v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4\}$ is also a basis of V.